

Test 1

Wednesday 21st, covers chapters 1 & 2

Ways functions might be special

$[F : A \rightarrow B]$

Name 1	Name 2	What makes them special
One-to-one function	Injection	Every element of the codomain has at most one pre-image.
Onto function	Surjection	Every element of the codomain has at least one pre-image (the range equals the codomain).
One-to-one correspondence	Bijection	Every element of the codomain has exactly one pre-image. (Both an injection and a surjection)

In terms of quantifiers (make table)

Property	Quantified version	Negation
Injection	$\forall x \in A \forall y \in A (x \neq y \Rightarrow f(x) \neq f(y))$ Equivalently: $\forall x \in A \forall y \in A (f(x) = f(y) \Rightarrow x = y)$	$\exists x \in A \exists y \in A (x \neq y \wedge f(x) = f(y))$
Surjection	$\forall y \in B \exists x \in A (f(x) = y)$	$\exists y \in B \forall x \in A (f(x) \neq y)$
Bijection	$\forall y \in B \exists! x \in A (f(x) = y)$ Here, $\exists!$ means “there exists exactly one.”	$\exists y \in B \forall x \in A (f(x) \neq y)$ or $\exists x \in A \exists z \in A (x \neq z \wedge f(x) = f(z))$ Some element has no pre-image, or some element has more than one pre-image.

Examples

Property	Example
Injection	Assigning IDs to students should be an injection: different students receive different IDs.
Surjection	Assigning cleaning responsibilities to volunteer teams should be a surjection: every responsibility is assigned to at least one team.

Bijections

They are central to §2.5 to show a particular function is not a bijection, either:

- Show it isn't an injection

or

- Show it isn't a surjection

Combining Functions

1. Via Arithmetics,

if we have 2 functions (same codomain) $f : A \rightarrow B$ and $g : A \rightarrow B$ (both f and g have same domain)

and we can apply arithmetic in the codomain (i.e $B = \mathbb{Q}$ or $B = \mathbb{N}$ or $B = \mathbb{R}$, etc.)

$f + g$ and fg

Are both functions from A to B defined by:

$$(f + g)(x) = f(x) + g(x)$$

$$(fg)(x) = f(x) \times g(x)$$

Example

If $f : \mathbb{N} \rightarrow \mathbb{R}$ is $f(x) = 3x + 5$

$g : \mathbb{N} \rightarrow \mathbb{R}$ is $g(x) = 2x + 12$

then

$$(f + g)(3) = f(3) + g(3) = 14 + 18 = 32$$

$$(fg)(3) = f(3) \times g(3) = 14 \times 18 = 252$$

2. Via composition

If $f : A \rightarrow B$ and $g : B \rightarrow C$

then we can use the output of f as the input of g

$g \circ f : A \rightarrow C$ given by $(g \circ f)(x) = g(f(x))$

Note:

f and g from the previous example, cannot be composed, because $\mathbb{R} \neq \mathbb{N}$

But if $f : \mathbb{N} \rightarrow \mathbb{N}$ is $f(x) = 3x + 5$ and

$g : \mathbb{N} \rightarrow \mathbb{N}$ is $g(x) = 2x + 12$ then

$$(g \circ f)(2) = g(f(2)) = g(11) = 33 \text{ also}$$

$$(f \circ g)(2) = f(g(2)) = f(16) = 53$$

Example:

P = set of people

B = set of bands

C = costs

we could have $f : P \rightarrow B$

$f(x)$ = x 's favorite band

$g : B \rightarrow C$ with $g(y)$ = cost of the cheapest ticket to see y

$(g \circ f)(\text{Eve})$ = The cost of a ticket to see Eve's favorite band

3. If $f : A \rightarrow B$ is a bijection

then for every $y \in B$ we can talk about the pre-image of y

In this case, we can define $f^{-1} : B \rightarrow A$

by $f^{-1}(y) = x$ iff $f(x) = y$

Example

If $g : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ given by $g(x) = x^2$ then

$$g^{-1}(y) = \sqrt{y}$$

i.e

$$\sqrt{y}^2 = y$$

If we'd use the same rule with different domain or codomain, we might not have an inverse.

Example:

$h : \mathbb{R} \rightarrow \mathbb{R}^+$ (given by $h(x) = x^2$)

is not a bijection because $h(-2) = h(2)$ (so we couldn't pick a correct value for $h^{-1}(4)$)

$l : \mathbb{R}^+ \rightarrow \mathbb{R}$ (given by $l(x) = x^2$)

is not a bijection because -1 has no pre-image

Example:

$f\{\text{anger, sadness, fear}\} \rightarrow \{\text{red, blue, yellow}\}$

given by

$f(\text{anger} = \text{red}) \quad f(\text{sadness} = \text{blue}) \quad f(\text{fear} = \text{yellow})$

is a bijection with domain $\neq$ codomain,

in this case $f^{-1}(\text{blue}) = \text{sadness}$

Properties

If $f : A \rightarrow B$ is a bijection
then

$$\forall x \in A f^{-1}(f(x)) = x$$

and

$$\forall y \in B f(f^{-1}(y)) = y$$

We'll talk about graphs of functions after reading week. (don't need them until chapter 9)

§2.4 Sequences and Summations

Definition:

A sequence is a function whose domain is a subset of $\mathbb{Z}$, (especially $\mathbb{N}$ or $\mathbb{Z}^+$)

Java arrays make great sequences.

In this case we often use subscript notation so if the sequence is $\{a_n\}$ then (we can be explicit on domain with $\{a_n\}_{n \in \mathbb{N}} \vee \{a_n\}_{n \in \mathbb{Z}^+}$ etc.)
 a_n is the image of n

Example:

cow, dog, cat, turtle, cow, cat, ...

- first instance of "cow" starts at 0, (image 0), then image 1 then 2, etc...

Sequences are special

because we can (completely) describe them with a (possible infinite) list of outputs,

Some nice sequences

Example:

7, 11, 15, 19, 23, 27, ... (called an arithmetic sequence)

the difference between consecutive terms is constant (in this case its 4)

Generically an arithmetic sequence has an n 'th term of $a + dn$

a is the initial value

d is the common difference

n is the index (which starts at 0)

in our case:

$$a = 7 \quad d = 4$$

Example:

1, 2, 4, 8, 16, ...

If each entry is $r \times$ the previous we get a geometric sequence

$$b_n = ar^n$$

a = initial value

r = common ratio

n = is the index